DR. B. C. ROY ENGINEERING COLLEGE, DURGAPUR

AN AUTONOMOUS INSTITUTE

AFFILIATED TO: MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL

Applied Exponential Models for Electrical Engineering I & II

EE-AOC-581 & EE-AOC-681

B.Tech (EE) | 5th and 6th Semester

20 Selected Students | Batch B2024–28

One Curve, Many Worlds: From Circuit Theory to Motor Drive

Compact Syllabus – Add-On Course EE-AOC-581 & EE-AOC-681

Prepared by:

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Department of Electrical Engineering

Academic Year 2026–27

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Contents

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1 Course Identification

Field	Details
Course Title	Applied Exponential Models for Electrical Engineering I & II
Course Tagline	*One Curve, Many Worlds: From Circuit Theory to Motor Drive*
Course Code	EE-AOC-581 & EE-AOC-681
Course Type	Add-On / Value-Added Course (Non-Credit, Certificated)
Target Programme	B.Tech (Electrical Engineering)
Target Semester	5th Semester (Part I) and 6th Semester (Part II)
Target Batch	B2024–28, Academic Year 2026–27
Batch Strength	20 students (selected by competitive process)
Total Contact Hours	74 Hours (Lecture + Hands-On, integrated)
Semester 5 – Part I	42 Hours
Semester 6 – Part II	32 Hours
Venue	Sim Lab-III, EE Dept., BCREC (Tesla Research Lab as extended venue when required)
Compliance	NBA (11 POs + 3 PSOs), NAAC 2.3/3.4, AICTE Model Curriculum

2 Prerequisites

2.1 Formal Subject Prerequisites (Completed by End of Semester 4)

Sl.	Subject	Concepts Relevant to This Course
1	Basic Electrical Engineering	RC/RL transient response, KVL/KCL, Thevenin/Norton, phasors
2	Network Theory	Mesh/node analysis, transient/steady-state, Laplace basics
3	Electromagnetic Fields	Wave propagation $e^{j\omega t}$, energy in fields
4	Electrical Machines – I	B–H curve, transformer equivalent, DC motor
5	Analog Electronics	Diode I–V characteristic, op-amp, feedback circuits
6	Measurement and Instrumentation	Lissajous figures on CRO, phase and frequency measurement

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2.2 Co-requisite Subjects (Running Concurrently in Semester 5)

Sl.	Subject	Connection Used in This Course
1	Electrical Machines – II	Induction motor torque–slip curve, PMSM basics
2	Power System – I	Transmission line sag (catenary), P–V nose curve
3	Control Systems	Root locus, Bode plot, pole-zero response, e^{st}
4	Power Electronics	Converter topologies, switching waveforms, duty cycle
5	EV and Hybrid Vehicles	Battery SoC curve, BESS integration

3 Student Selection and Course Rules

3.1 Selection Procedure

Students are selected through three stages conducted before the start of Semester 5.

Stage	Mode	Platform	Target	Description
1	Expression of Interest	Google Form	Up to 40	Motivation, time availability, prior exposure to Python/C and circuits. No elimination at this stage.
2	Aptitude Quiz	Google Quiz (30 min)	Up to 30	Basic Python/C concepts and fundamental electrical concepts. Minimum qualifying score: 60%.
3	Technical Viva Voce	In-person, EE Dept.	Final 20	10-minute interaction assessing curiosity and baseline understanding. Panel: Course Instructor and HoD EE.

Timeline: Stage 1 opens 4 weeks before Semester 5 starts. Stage 2 is held 2 weeks before. Stage 3 is held 1 week before. The batch list is frozen on Day 1 of Semester 5. No additions are permitted after the first session.

3.2 Course Rules (Brief)

1. **Attendance:** Minimum 80% of all scheduled contact hours is mandatory. College-declared and government gazette holidays are not counted as contact hours and do not affect attendance calculation. Any other absence – including collective absence by a group of students – is marked absent without exception. There is no provision for make-up sessions or attendance relaxation on any grounds of group absence. Late entry beyond 10 minutes counts as absent.

2. **Continuation to Semester 6:** Only students with minimum 50% aggregate and at least 80% attendance in Semester 5 (Part I) are eligible to continue.

3. **Certificate:** This is a non-credit course. It does not appear in the MAKAUT mark sheet. A departmental certificate of completion is issued on passing both semesters.

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4. **Lab notebook:** Submissions must include handwritten circuit diagrams and sketches with Python/Scilab printouts attached. Fully printed submissions are not accepted.
5. **Plagiarism:** Zero marks for the affected module and a note in the departmental record.
6. **Equipment:** Damage to Sim Lab-III equipment due to negligence is the student's responsibility.
7. **Withdrawal:** Students who withdraw after Stage 3 selection without one week's prior notice forfeit the right to rejoin a future batch.

4 Course Content

Module	Content	Hours
	Semester 5 – Part I (42 Hours)	
1	**Network Theorems and Energy Properties of R, L, C** Thevenin/Norton in real-world contexts; R, L, C impedance vs. frequency; Volt-second balance of L and Current-second balance of C; Euler's formula and the complex plane; impedance locus; pole and zero as complex-plane objects.	8
2	**The Exponential Family – One Curve, Many Worlds** Pure growth e^t and decay e^{-t}; saturating rise $(1 - e^{-t})$; sigmoid; hyperbolic tangent tanh (B–H curve connection); catenary cosh (transmission line sag connection); damped oscillation $e^{(\sigma+j\omega)t}$; unification session – all curves on one plot with pole-location interpretation.	8
3	**Linear Voltage Regulator – Design, Protection, and Fabrication** Datasheet reading (7805, LM317); fixed regulator (7805/7812/7905); variable regulator (LM317/LM337); protection circuits (current limiting, thermal shutdown, overvoltage crowbar); complete bench power supply integration.	6
4	**Switch Mode Power Supply – Buck and Boost (Open Loop)** Inductor current waveforms; $V_o/V_{in} = D$ (Buck) and $1/(1-D)$ (Boost) from Volt-second balance; state-space averaging; small-signal model; control-to-output transfer function $G_{vd}(s)$; Bode plot; ESR zero; right-half-plane zero; non-minimum phase response of Boost.	10

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Module	Content	Hours
5	**Gate Pulse Generation – From Analog to Professional IC** 555 timer and comparator-based PWM; Arduino Timer-register PWM; optical isolation (PC817, 6N137); TI UCC27524 dual 5A gate driver (datasheet reading, R_G design, UVLO, dead time, Boost chopper application); ESP32 LEDC peripheral for precision PWM.	10
	Semester 5 Sub-total	**42**
	Semester 6 – Part II (32 Hours)	
6	**Closed-Loop Control – Type-II and Type-III Compensator Design** Review of Module 4 Bode plots; Type-I compensator and its limits; Type-II (one zero, two poles): K-factor method, component calculation, op-amp realisation; Type-III (two zeros, three poles): K-factor extended, ESR zero compensation; analog and digital (Arduino/ESP32) implementation; loop gain verification.	10
7	**Generalised Machine Theory and dq0 Transformation for Drive Control** DC motor as naturally separated machine; why AC machine control is hard; generalised machine theory (three equations: voltage, energy, torque); reference frame theory; Park abc-to-dq0 transformation (d-axis = flux, q-axis = torque); validation on DC motor; application to BLDC and PMSM ($i_d = 0$ strategy, MTPA); FOC block diagram (Type-II current controllers from Module 6 identified explicitly); course convergence session.	12
8	**Real-World Validation – OpenModelica and Hardware-in-the-Loop** OpenModelica environment; RC/RL verification (Module 1 baseline); Buck/Boost open-loop model; closed-loop converter model (Module 6 compensator); PMSM dq model with FOC; HIL Option A: controller on Arduino/ESP32, plant in OpenModelica; PIL Option B: controller in simulation, real converter as plant; final assessed demonstration.	10
	Semester 6 Sub-total	**32**
	Grand Total	**74**

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5 Course Outcomes (COs)

Taxonomy: Anderson and Krathwohl Revised Bloom's Taxonomy. Abbreviations: **Re** = Remember, **Un** = Understand, **Ap** = Apply, **An** = Analyse, **Ev** = Evaluate, **Cr** = Create.

CO	Statement	Bloom Level
CO1	**Identify** and **classify** the exponential function family (e^t, e^{-t}, $1 - e^{-t}$, sigmoid, tanh, cosh, $e^{j\omega t}$, $e^{(\sigma+j\omega)t}$) and **correlate** each member to its physical phenomenon in electrical circuits, power systems, and machines.	BL1 (Re), BL2 (Un)
CO2	**Analyse** the steady-state and transient behaviour of R, L, C networks and Buck/Boost converter circuits using Volt-second and Current-second balance, state-space modelling, and Bode plot interpretation including ESR zero and non-minimum phase response.	BL4 (An)
CO3	**Apply** datasheet reading skills and circuit design principles to **implement** linear voltage regulators (fixed and variable), protection circuits, and gate drive circuits (analog, Arduino/ESP32, optically isolated, UCC27524-based).	BL3 (Ap), BL6 (Cr)
CO4	**Design** Type-II and Type-III compensators for closed-loop Buck/Boost control using the K-factor method and **evaluate** the achieved stability margins through simulation and hardware verification.	BL5 (Ev), BL6 (Cr)
CO5	**Explain** the physical necessity of the abc-to-dq0 (Park) transformation for AC machine control and **apply** field-oriented control (FOC) principles to DC, BLDC, and PMSM drives.	BL2 (Un), BL3 (Ap)
CO6	**Build** and **evaluate** a complete motor drive system from gate pulse generation to closed-loop speed control using OpenModelica simulation and/or HIL/PIL implementation.	BL5 (Ev), BL6 (Cr)

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6 Graduate Attributes

The following graduate attributes are explicitly developed through this course and map to NBA Graduate Attributes for B.Tech programmes.

Sl.	Graduate Attribute	How Developed	Modules
1	Mathematical Thinking	Exponential family unification; state-space derivation; Park transformation; K-factor compensator design – structured mathematical reasoning throughout, not formula substitution.	M1, M2, M4, M6, M7
2	Computational Modelling	Python dashboards; Scilab/Xcos; LTspice; OpenModelica; HIL/PIL – students build and interpret computational models at every stage of the course.	M2, M4, M6, M7, M8
3	Engineering Judgement	Choosing Type-II vs. Type-III compensator; selecting crossover frequency; analog vs. digital implementation decision; interpreting non-minimum phase response – real consequences on hardware.	M4, M5, M6, M7
4	Problem Solving	Every module follows a problem-solution narrative. Module 4 identifies the control problem; Module 6 solves it. The course structure itself models systematic engineering problem framing.	All Modules
5	Analytical Reasoning	Bode plot interpretation; pole-zero physical meaning; dq0 validation on DC motor; viva questions require first-principles reasoning, never result recall.	M1, M2, M4, M6, M7

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7 CO – PO – PSO Mapping

As per current NBA criteria for B.Tech programmes: 11 Programme Outcomes (PO1–PO11) and 3 Programme Specific Outcomes (PSO1–PSO3). Mapping strength: **3** = High, **2** = Moderate, **1** = Low, – = No mapping.

PO Definitions: PO1 Engineering Knowledge, PO2 Problem Analysis, PO3 Design/Development of Solutions, PO4 Conduct Investigations, PO5 Modern Tool Usage, PO6 Engineer and Society, PO7 Environment and Sustainability, PO8 Ethics, PO9 Individual and Team Work, PO10 Communication, PO11 Project Management and Finance.

PSO Definitions (B.Tech EE, BCREC): PSO1 Apply knowledge of machines, power systems, and power electronics to analyse and design electrical energy systems. PSO2 Use simulation and embedded tools to implement and validate EE solutions. PSO3 Demonstrate awareness of energy efficiency, sustainability, and emerging technologies.

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	**3**	**2**	**1**	**2**	**2**	–	–	**1**	–	**2**	–	**3**	**1**	**1**
CO2	**3**	**3**	**2**	**3**	**3**	–	–	**1**	–	**1**	–	**3**	**3**	**1**
CO3	**2**	**2**	**3**	**2**	**3**	**1**	–	**2**	**2**	**2**	**1**	**3**	**3**	**1**
CO4	**3**	**3**	**3**	**3**	**3**	–	–	**1**	**1**	**2**	**2**	**3**	**3**	**1**
CO5	**3**	**3**	**2**	**2**	**3**	**1**	–	**1**	–	**2**	–	**3**	**3**	**2**
CO6	**3**	**3**	**3**	**3**	**3**	**1**	**1**	**2**	**2**	**2**	**2**	**3**	**3**	**2**
Avg.	**2.83**	**2.67**	**2.33**	**2.50**	**2.83**	**0.33**	**0.17**	**0.83**	**0.50**	**1.50**	**0.50**	**3.00**	**2.67**	**0.67**

Strongly mapped: PO1 (Engineering Knowledge), PO2 (Problem Analysis), PO3 (Design), PO5 (Modern Tool Usage). PSO1 and PSO2 mapped at high strength across all COs.

8 Assessment Structure

Philosophy: This course carries no written examination. Understanding is demonstrated by doing. Assessment is continuous, practical, and oral.

Sl.	Component	Weight	Description
1	Lab Notebook / Module Report	30%	Submitted after each module. Must include circuit diagrams (handwritten), waveform sketches, Python/Scilab printouts, annotated datasheets, and one physical interpretation paragraph per result. This serves as the primary CO attainment evidence.
2	Practical Demonstration (Mini Project)	20%	Conducted at the end of each semester. The student builds a specified circuit or runs a specified simulation, operates it, and explains every design choice and observed result before a faculty panel.
3	Viva Voce	10%	10–15 minute oral session per student at the end of each semester. Questions target physical understanding. Example: "Why does the Boost converter output dip before rising?"
4	End semester open book examination	40%	End semester an open book examination (No Provision of re-examination)

Passing Criterion: Minimum 50% aggregate and minimum 80% attendance (college/gazette holidays excluded; collective absence carries no special status). A departmental certificate of completion is issued on passing both semesters.

9 References and Software

9.1 Textbooks and Technical Documents

1. N. Mohan, T. M. Undeland, and W. P. Robbins, *Power Electronics: Converters, Applications, and Design*, 3rd ed., Wiley, 2002.

2. P. C. Krause, O. Wasynczuk, and S. D. Sudhoff, *Analysis of Electric Machinery and Drive Systems*, 3rd ed., IEEE Press/Wiley, 2013.

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3. D. W. Hart, *Power Electronics*, McGraw-Hill, 2011.

4. Texas Instruments Application Report SLVA301: *Compensator Design Procedure for Buck Converter with the TPS40200*.

5. D. Venable, "The K-Factor: A New Mathematical Tool for Stability Analysis and Synthesis," *Proc. Powercon 10*, 1983.

9.2 Online Resources and Datasheets

1. TI UCC27524 Datasheet: https://www.ti.com/product/UCC27524
2. TI WEBENCH Power Designer (free, online): https://webench.ti.com
3. OpenModelica Documentation: https://openmodelica.org/doc/OpenModelicaUsersGuide/
4. Python Control Systems Library: https://python-control.readthedocs.io

9.3 Software Used in This Course (All Free and Open Source)

Sl.	Tool	Purpose
1	Python 3 (NumPy, SciPy, Matplotlib, python-control)	Curve plotting, Bode analysis, state-space, interactive dashboards
2	Scilab / Xcos	Signal flow and block diagram simulation
3	LTspice XVII (Free, Analog Devices)	SMPS circuit simulation (Modules 4 and 6)
4	OpenModelica	System-level simulation and HIL (Module 8)
5	BLDCSim	BLDC-specific motor simulation (Modules 7 and 8)
6	Arduino IDE	Arduino programming (Modules 5 and 6)
7	Arduino-ESP32 / ESP-IDF	ESP32 PWM and IoT integration (Module 5)
8	TI WEBENCH Power Designer (Free, online)	Type-II/III compensator cross-verification (Module 6)

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10 Course Prepared By

Field	Details
Name	Kingsuk Majumdar, Ph.D. (Engg.)
Designation	Assistant Professor (Grade II)
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Qualification	Ph.D., Electrical Engineering, NIT Durgapur (April 2023)
Teaching Experience	13+ years
Research Areas	Power systems, meta-heuristic optimisation, SMPS, BESS, motor drive, embedded systems
Official Email	kingsuk.majumdar@bcrec.ac.in
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GitHub	https://github.com/KingsukMajumdar
Research Lab	Tesla Research Lab (Departmental Research Lab, EE Dept., BCREC)[founder & Lab In-charge]
Course Design	MOOCs preferred basket for B. Tech (EE) honours (using NPTEL and Spoken Tutorial IIT Bombay) and serve as Dy. SPOC of MOOCs, BCREC since 2019
Key Publications	7 SCI/ESCI journal papers; 10 international conference papers; Indian Patent No. 536659
Design Registrations	UK Design No. 6508309 (Solar-Powered EV Charging Dock); UK Design No. 6520200 (Green Energy Charging Station)
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End of Syllabus: EE-AOC-581 & EE-AOC-681

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